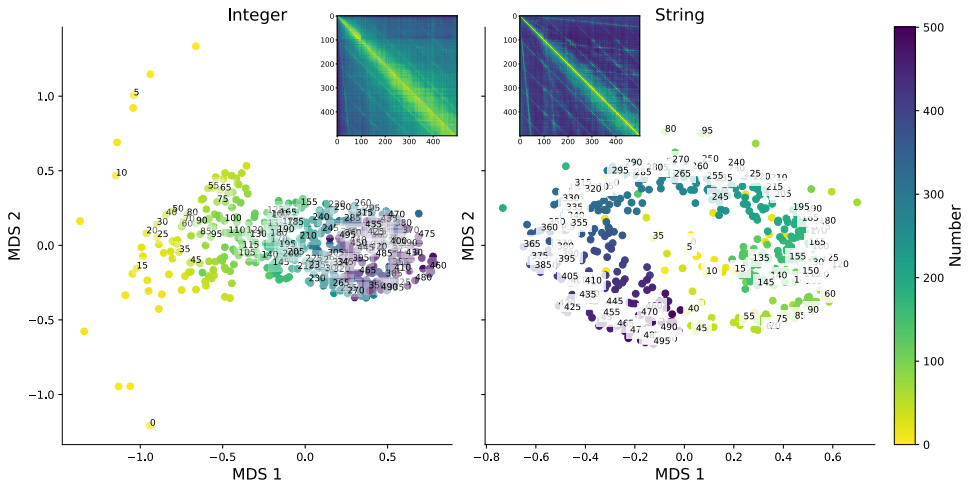

Figure 4. Decoded string and integer subspaces from Llama-3.1-8b using linear probes (see Methodology). The decoded similarity matrices are provided as insets along with their multidimensional scaling solutions (MDS). Integers are labeled every 5 points.

correlation is .917 with the Log-Linear distance and .393 with the Levenshtein distance (c.f., Table 2 in Appendix D).

To further highlight the structure of the latent subspaces, in Figure 4 we apply multidimensional scaling (Shepard, 1980) to the decoded similarity matrices in Figure 4 to derive two-dimensional maps of the relations between integers. We see that the probes nicely dissociate the integer and string subspaces. In the integer subspace, numbers are organized on a scale that captures numerical distance. On the other hand, in the string subspace, the scale is replaced with a non-linear pattern that closely follows edit distance.

5.3. Behavioral Implications

In the final experiment we wanted to see whether there are behavioral implications for the number-string tension in a naturalistic scenario. To do that, we presented the models with an empirical setting in which they required a test tube with a certain unavailable compound concentration (given in parts per million units; ppm). The models were asked to choose from two available test tubes the one containing the most similar concentration to the one desired. Crucially, we constructed diagnostic concentration triplets that lead to divergent answers depending on whether a Levenshtein distance is used or a Log-Linear one (see Methodology). Here is an example prompt: 'You require a compound with a concentration of approximately 785 ppm. Two test tubes are available: one containing 685 ppm and the other 791 ppm. Your task is to determine which test tube provides the most similar concentration to your required dosage.' Clearly, a concentration of 791 ppm is much more similar to 785 ppm than 685 ppm, but a model with strong string bias may erroneously choose 685 ppm which shares more digits with 785 ppm. We then evaluated the percentage of incorrect (Levenshtein-consistent) answers as a measure of string bias. In addition to the 3-digit triplets, we also considered 5-digit ones to see whether the problem may be exacerbated by the inclusion of more digits (e.g., 22565, 12565, and 28743; see Methodology). We also considered both orders of presentation to see whether there are any ordering effects when the Levenshtein-aligned option is presented first vs. the Log-Linear one ('Reverse' setting). The results are shown in Figure 5. We found that all models had some degree of string-bias errors but the range varied greatly and was enhanced for longer numbers. In the 3-digit case the largest percentage (averaged over order) was 36.86% for Llama-3.1-8b (followed by 11.38% for Llama-3.1-70b), and the smallest was 0.02% for GPT-4o. Likewise, in the 5-digit case, the largest percentage was 47.03% for Llama-3.1-8b (followed by 19.04% for Llama-3.1-70b), and the smallest was 0.41% for GPT-4o. Interestingly, the Llama-3.1 models exhibited substantially larger degrees of bias, as well as some degree of order effects in some cases (note that this cannot be attributed to a generic presentation bias since in that case we wouldn't expect to see a difference between the 3-digit and 5-digit scenarios). Overall, these results suggest that the number-string tension can manifest itself in a realistic context, and that the extent that such a context can suppress the stringiness behavior varies across models.

6. Discussion

We have provided evidence for a representational tension in large language models that learn to represent numbers by processing text input. This tension is reflected externally (through prompt-elicited similarity judgments), internally (through embedding probes), and could also propagate into a quantitative decision scenario.

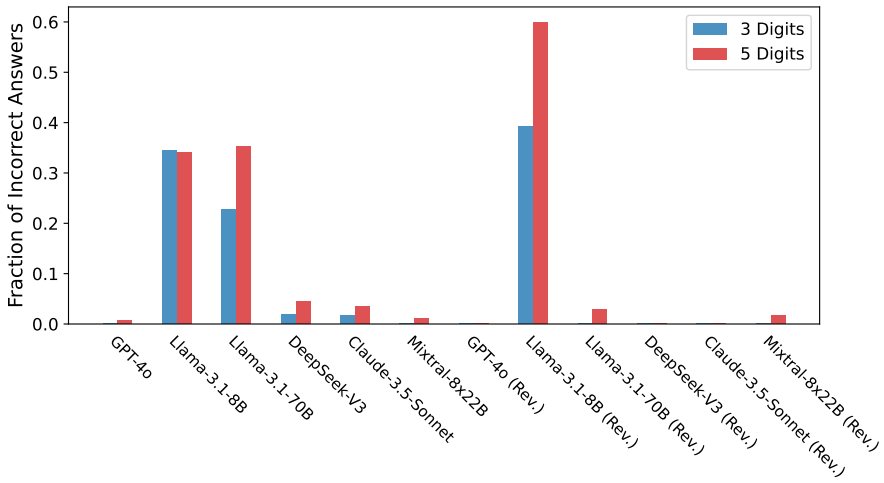

Figure 5. Probing string-bias in a naturalistic decision scenario. Bar plots indicate the faction of times an incorrect (Levenshtein-aligned) option was chosen for the 3-digit and 5-digit scenarios considered, and the two possible presentation orders (see Methodology). '(Rev.)' indicates the case in which the Levenshtein-aligned option was presented second (see prompt in Appendix A).

Our results suggest that the tension may arise from ambiguity in the 'type' of digit tokens in a given context. Specifically, by explicitly specifying the type of the tokens in the prompt (i.e., `int()` vs. `str()`) we showed how the similarity matrices in the default case can be pushed to align more strictly with string or log-linear distance. This ambiguity is reminiscent of polysemy and homonymy in language (Vicente & Falkum, 2017) whereby a single word can have multiple meanings, but in our case the ambiguity is extended across different systems. This entanglement of types is also reflected in our probing results whereby the linearly decoded similarity matrices had some residual shared structure.

Our findings are consistent with recent probing work showing that LLMs are able to encode the value of numbers (Zhu et al., 2025). Interestingly, however, we found that when the setting is not strictly mathematical as in arithmetic, LLMs appear to exhibit a more psychologically plausible log-linear representation, though blended with string distance. This is consistent with recent work showing that LLMs encode rich sensory knowledge (Marjieh et al., 2024b). The results also make contact with the literature on how LLMs perform numerical operations (Hanna et al., 2023; Stolfo et al., 2023) and the possible sources of numerical errors that could arise from their mode of learning (McCoy et al., 2024a;b). One possibility is that LLMs attempt to categorize the type of an integer token from context, and then use that to specify a set of operations on units of that type. For example, if an LLM views the numbers 1 and 2 as strings, then asking it to sum the two numbers may result in 12 rather than 3. Future work could interrogate whether there is indeed a causal link between such type-categorization mechanisms and numerical errors.

We end by discussing some limitations that point towards future directions. First, due to resource limitations our LLM results were restricted to zero temperature (though see Appendix F) and the probing analysis was limited to the smaller Llama-3.1-8b model. Future work could look into how the results generalize to a stochastic sampling setting and other models for probing. Second, while we explored the effect of context on number similarity judgments, it remains to be seen whether one could causally intervene at the embedding level to steer the model from one token type to another. Third, while our naturalistic decision scenario was designed to be as diagnostic as possible, further work is necessary to assess how pervasive this issue is in more generic daily numerical use cases. Relatedly, reasoning-based models are increasingly becoming the *de facto* standard in user-facing applications (e.g., OpenAI's o1 Jaech et al., 2024, DeepSeek's R1 DeepSeek-AI, 2025). Extending our findings to these models is a promising avenue for future work, as their ability to reason may fundamentally alter how similarity judgments are produced. Finally, our work was restricted to the domain of numbers but our results may be a special case of a more broad symbol-string duality that impacts other domains (e.g., code and special characters). We hope to report on these directions in the future.

As artificial intelligence becomes increasingly intermingled with human intelligence, knowing what a number is becomes an increasingly important concern. McCulloch (1961) argued that the human capacity to understand numbers could be acquired by machines based on artificial neural networks. Our results suggest that there are still meaningful differences in how humans and large language models based on artificial neural networks conceive of numbers.

Reproducibility. Code for reproducing the analyses can be accessed here: https://github.com/vminvsky/numbers-in-llms

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